Chapter 3

Data and Signals

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Note

To be transmitted, data must be transformed to electromagnetic signals.

3-1 ANALOG AND DIGITAL

Data can be analog or digital. The term analog data refers to information that is continuous; digital data refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.

Topics discussed in this section:

- **Analog and Digital Data**
- **Analog and Digital Signals**
- **Periodic and Nonperiodic Signals**

Analog and Digital Data

- Data can be analog or digital.
- “**Analog data**”
  - refers to information that is continuous
  - take on continuous values.
- “**Digital data**”
  - refers to information that has discrete states
  - take on discrete values.

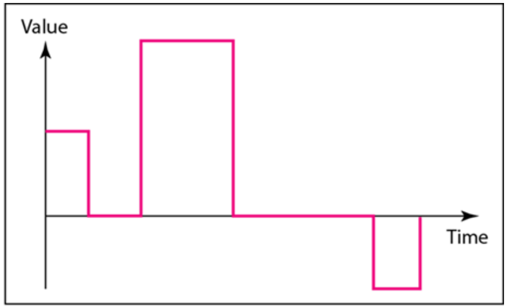
Analog and Digital Data

- **Example**:
  - An analog clock that has hour, minute, and second hands gives information in a continuous form. The movements of the hands are continuous.
  - On the other hand, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.

Analog and Digital Signals

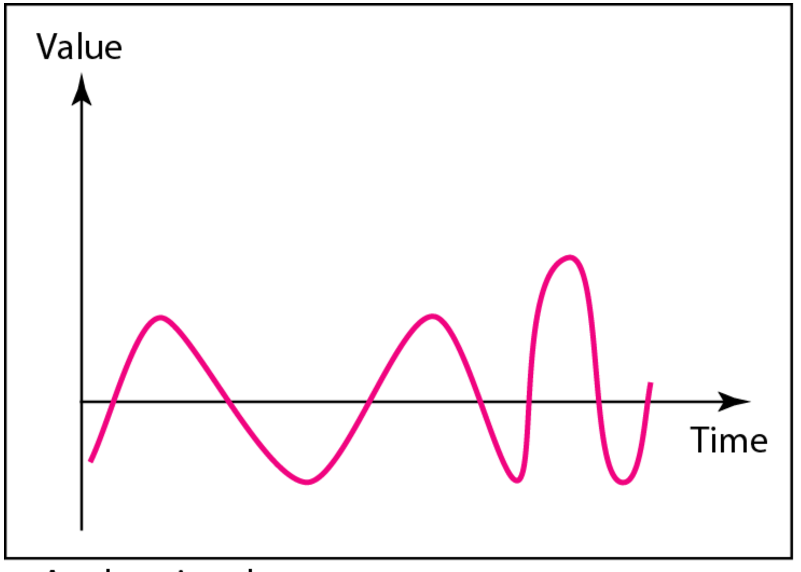
- Like the data they represent, signals can be analog or digital.
- An "**Analog signal**" has infinitely many levels of intensity in a period of time.
- "**Digital signals**" can have only a limited number of defined values.

a. Analog signal

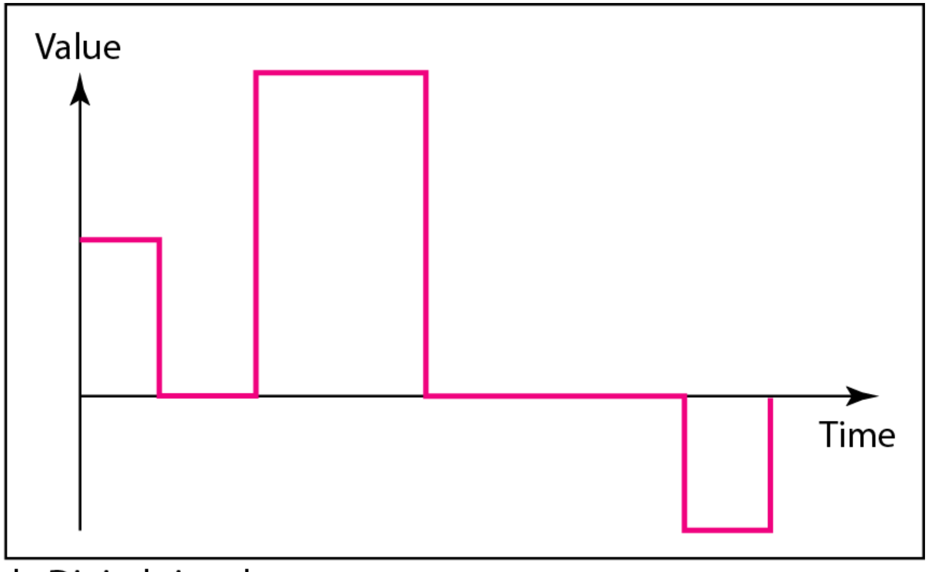


b. Digital signal

Figure 3.1 *Comparison of analog and digital signals*



a. Analog signal



b. Digital signal

Periodic and Nonperiodic

- Both analog and digital signals can be periodic or nonperiodic(or aperiodic)

- A **periodic signal** completes a pattern within a measurable time frame, called a **period**, and repeats that pattern over subsequent identical periods.
- The completion of one full pattern is called a **cycle**.

- A **nonperiodic signal** changes WITHOUT exhibiting a pattern or cycle that repeats over time.

Note

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

3-2 PERIODIC ANALOG SIGNALS

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.
Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

Topics discussed in this section:

- **Sine Wave**
- **Wavelength**
- **Time and Frequency Domain**
- **Composite Signals**
- **Bandwidth**

Periodic Analog Signals

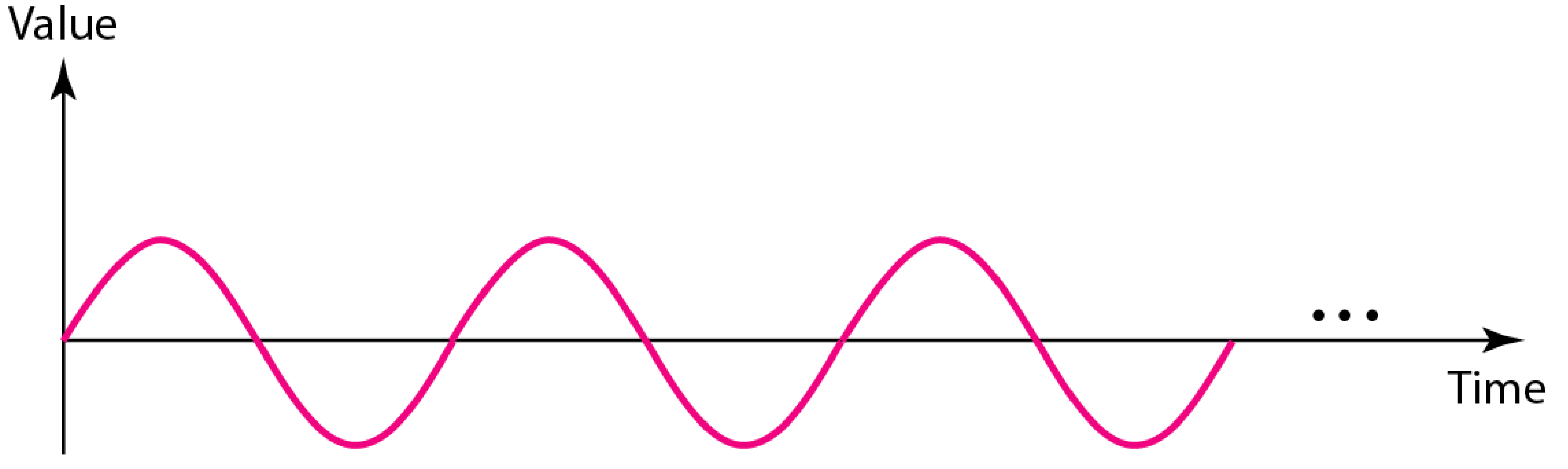
- **Periodic Analog Signals**

  A. simple – a sine wave
  B. composite - composed of multiple sine waves

Sine Wave

- the most **fundamental form** of a periodic analog signal.
- fully described by 3 parameters:
  - the peak amplitude,
  - the frequency, and
  - the phase.

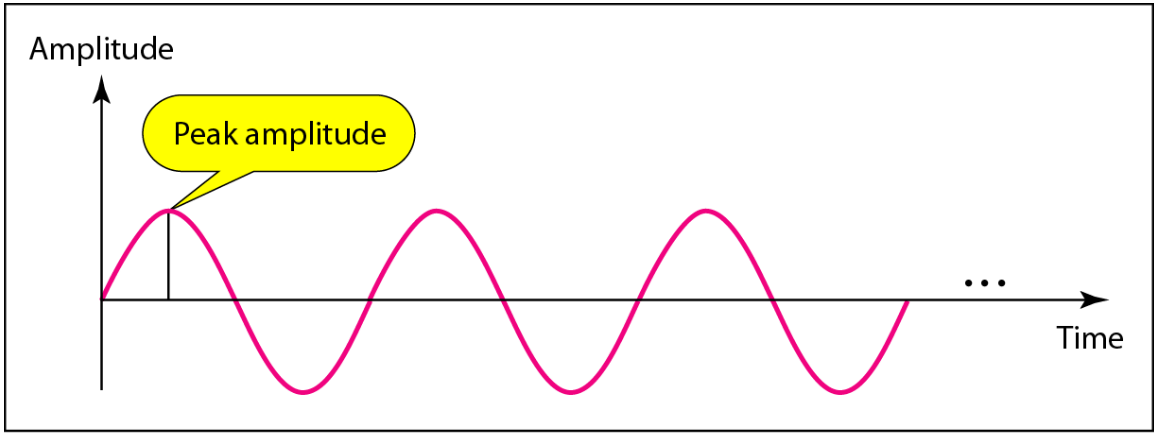
Figure 3.2 *A sine wave*



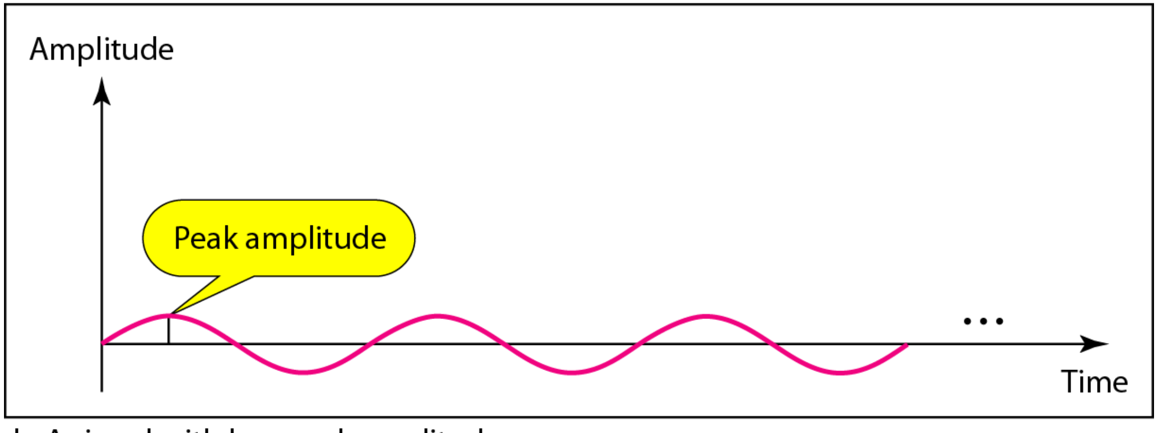
Sine Wave

- **Peak Amplitude:**
  - is the absolute value of its **highest intensity**, proportional to the energy it carries.
  - For electric signals, it is normally measured in **volts (V)**.
  - **Example**: The power in your house can be represented by a sine wave with a peak amplitude of **325 V**. However, its rms value is, 1/(2^½) X 325 = 0.707 X 325 = **230 V**.

Figure 3.3 *Two signals with the same phase and frequency, but different amplitudes*



a. A signal with high peak amplitude



b. A signal with low peak amplitude

Period and Frequency

- **Period** **(**in seconds**)** refers to ***the amount of time a signal needs to complete 1 cycle***.
- Period is formally expressed in seconds.

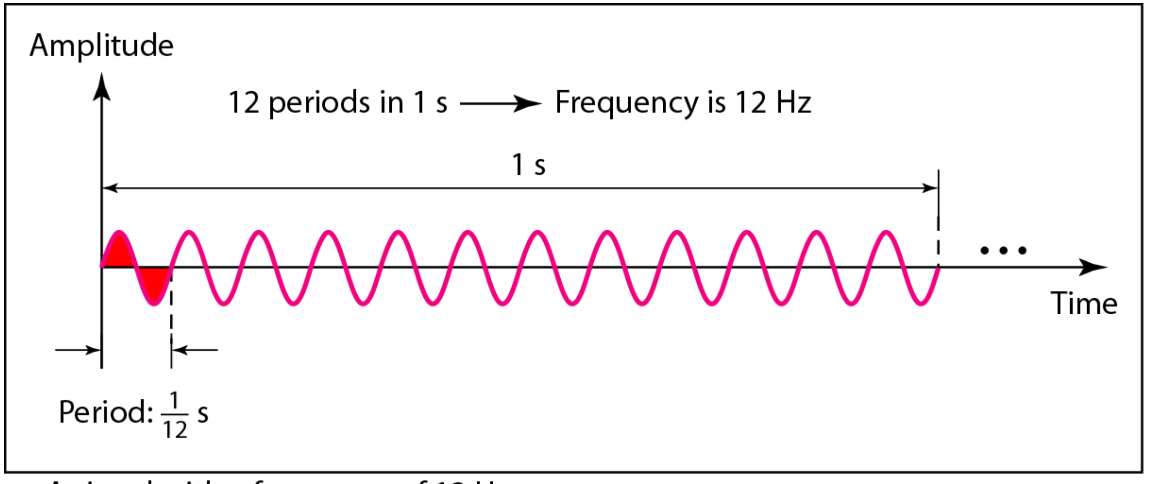
- **Frequency** refers **to** ***the number of periods in 1s.***
- Frequency is formally expressed in Hertz (Hz)

Note

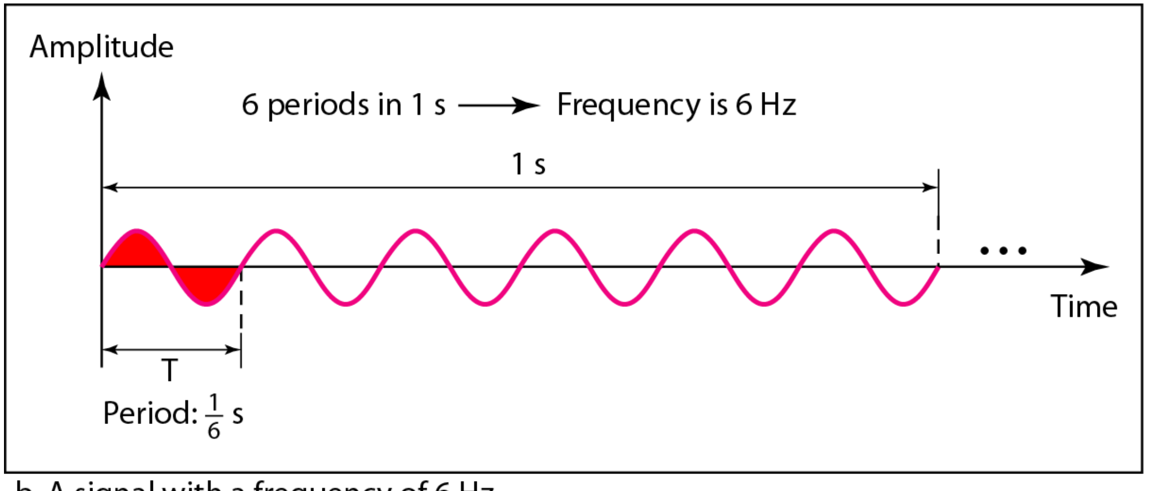
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \qquad \text{and} \qquad T = \frac{1}{f}$$

Figure 3.4 *Two signals with the same amplitude and phase, but different frequencies*



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Table 3.1 ***Units of period and frequency***

Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Example 3.1

*The power used at home (USA) has a frequency of **60 Hz**. The period of this sine wave can be determined as follows:*

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

Calculate for India!

Example 3.2

The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution

First, we change 100 ms to seconds, and then we calculate the frequency from the period (1 Hz = 10^{-3} kHz).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

Note

If a signal does not change at all (period = ∞), its frequency is zero.

If a signal changes instantaneously (period = 0), its frequency is ∞.

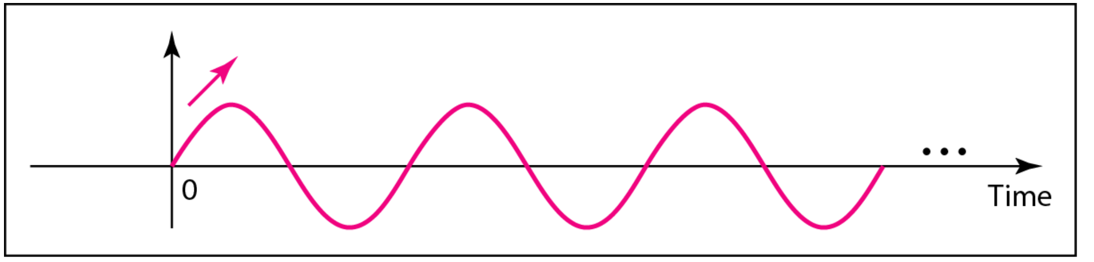
Phase

- The term **phase**, or phase shift, describes the position of the waveform relative to time 0.
- If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift.
- Phase is measured in degrees or radians [360° is 2π rad; 1° is $2\pi/360$ rad, and 1 rad is $360/(2\pi)\ ^\circ$].

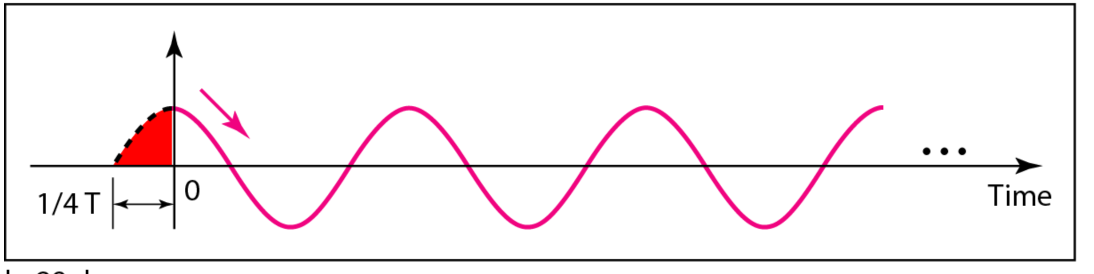
Note

Phase describes the position of the waveform relative to time 0.

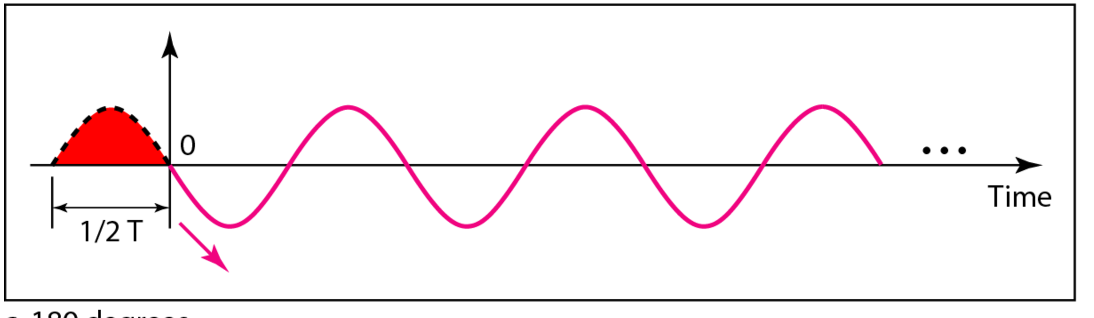
Figure 3.5 *Three sine waves with the same amplitude and frequency, but different phases*



a. 0 degrees



b. 90 degrees



c. 180 degrees

Example 3.3

A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

Solution

We know that 1 complete cycle is 360°. Therefore, 1/6 cycle is

$$\boxed{\frac{1}{6} \times 360 = 60^\circ = 60 \times \frac{2\pi}{360}\ \text{rad} = \frac{\pi}{3}\ \text{rad} = 1.046\ \text{rad}}$$

Wavelength

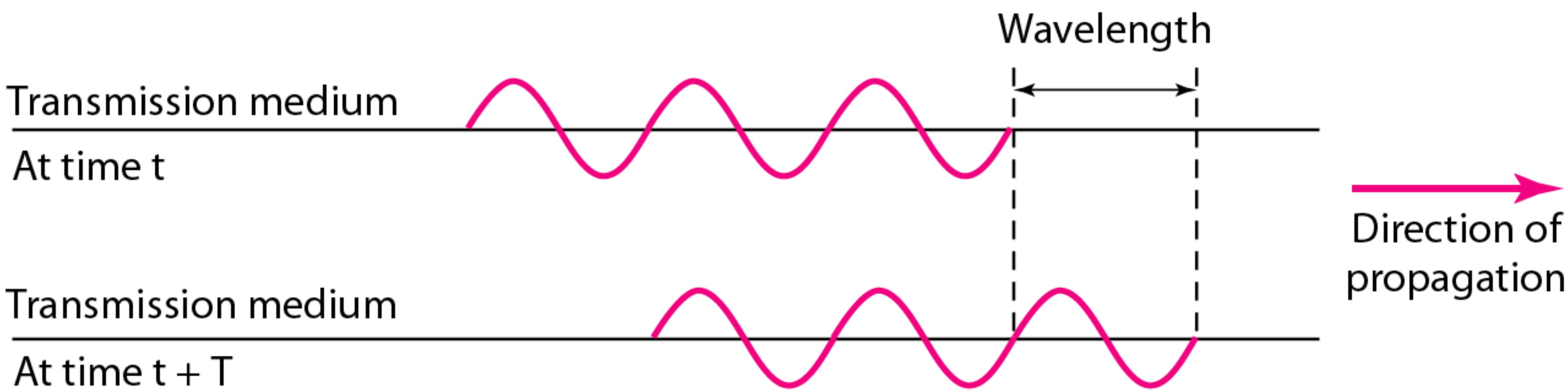
- The **wavelength** is the **distance** a simple signal can travel in one period.

$$\textit{Wavelength} = (\textit{propogation speed}) \times \textit{period} = \frac{\textit{propogation speed}}{\textit{frequency}}$$

$$\therefore \lambda = \frac{c}{f}$$

- The propagation speed of electromagnetic signals depends on the medium and the signal frequency (for red light 4X10^14).
- For **example**, in a vacuum, light is propagated with a speed of 3 × 10^8 m/s. That speed is lower in air and even lower in cable.
- Hence, the wavelength is shorter in air and even shorter in cable compared to vacuum.

Figure 3.6 *Wavelength and period*

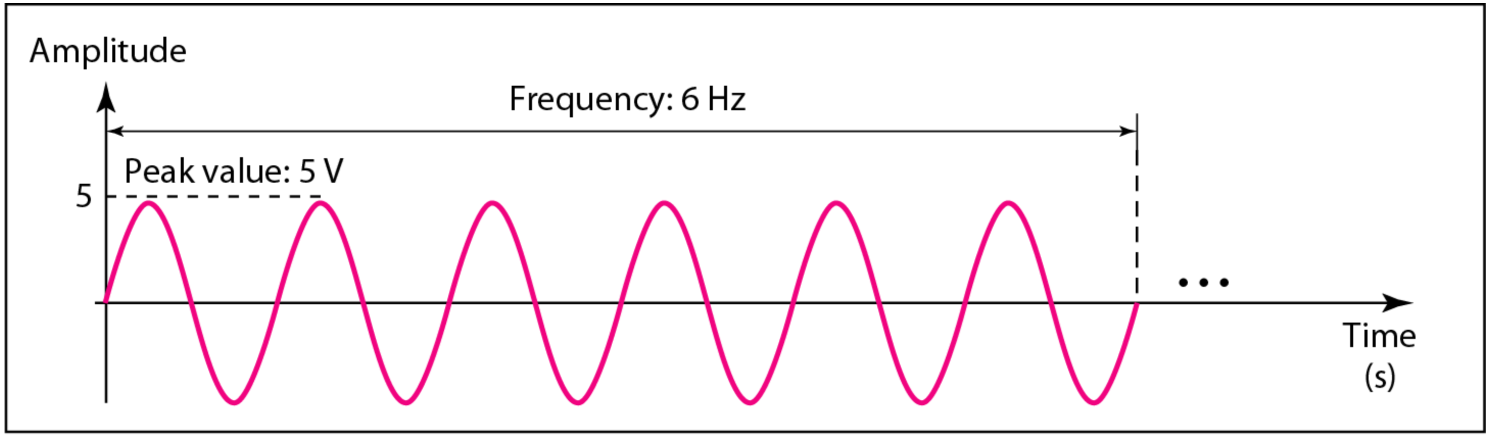


Time and Frequency Domains

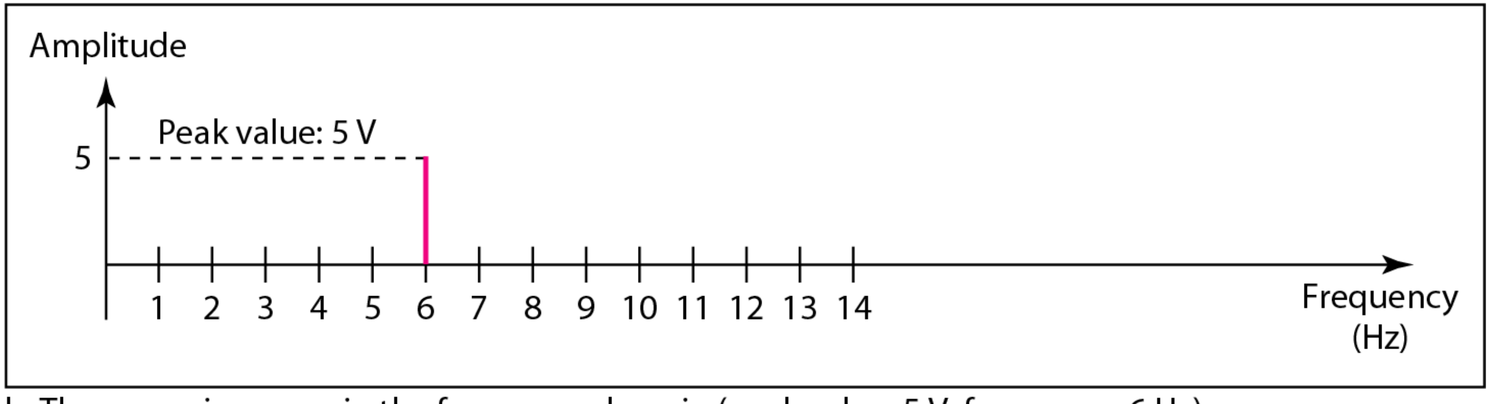
- **Time-domain plot** of the sine wave shows changes in signal amplitude w.r.t. time.

- **Frequency-domain plot** of the sine wave shows the relationship between amplitude and frequency.
- A frequency-domain plot is concerned with only the peak value and the frequency.

Figure 3.7 *The time-domain and frequency-domain plots of a sine wave*



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

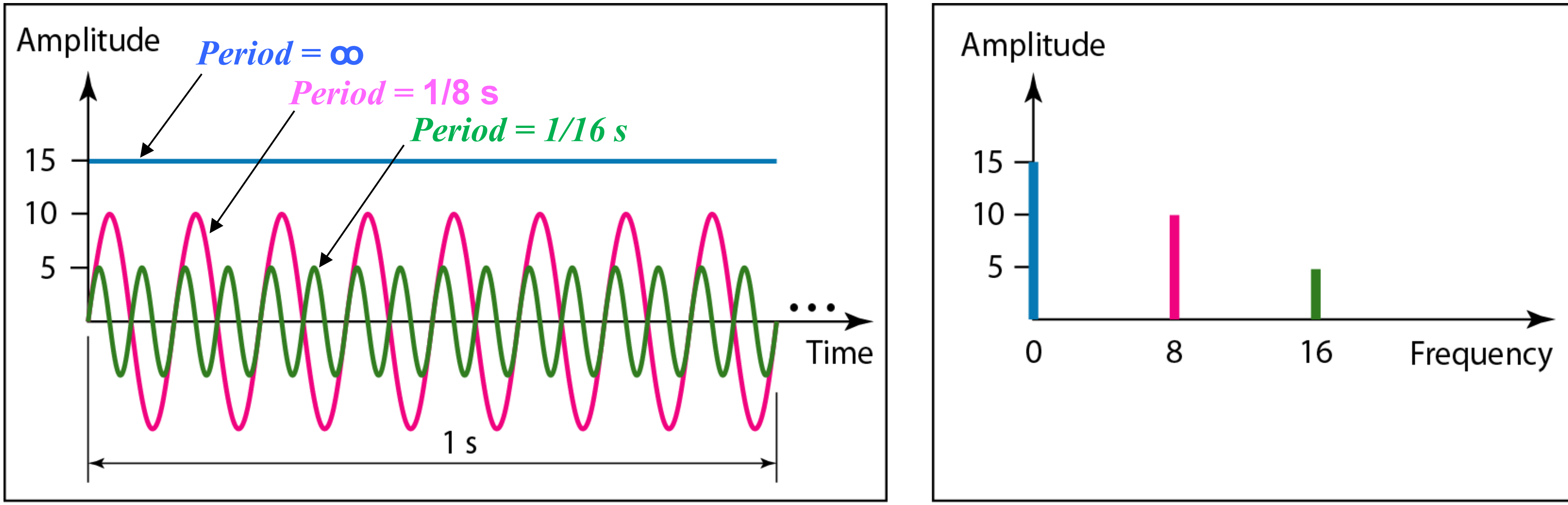
Note

A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.8 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Figure 3.8 *The time domain and frequency domain of three sine waves*



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

b. Frequency-domain representation of the same three signals

Signals and Communication

- A single-frequency sine wave is NOT useful in data communications
- We need to send a **composite signal**, a signal made of many simple sine waves.
- As you know from EM, according to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

Composite Signals and Periodicity

- If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies.
- If the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.

Example 3.4

Figure 3.9 shows a periodic composite signal with frequency f. This type of signal is not typical of those found in data communications. We can consider it to be three alarm systems, each with a different frequency. The analysis of this signal can give us a good understanding of how to decompose signals.

Figure 3.9 *A composite periodic signal*

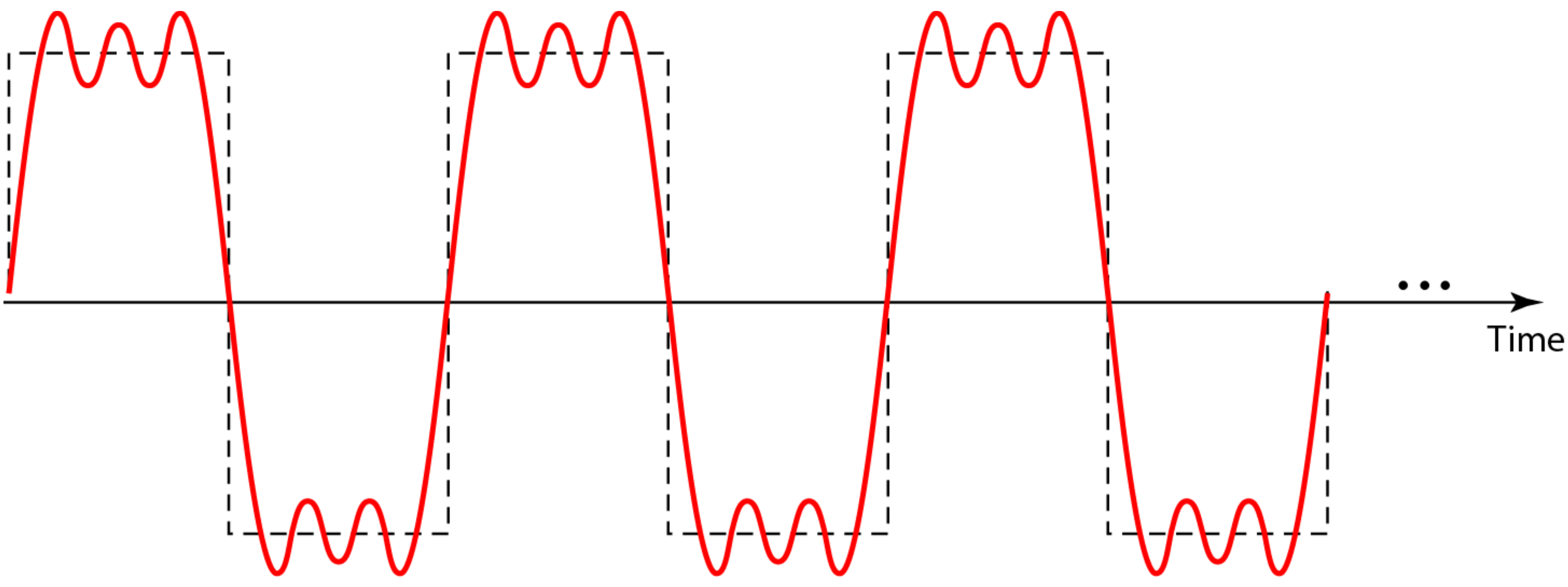
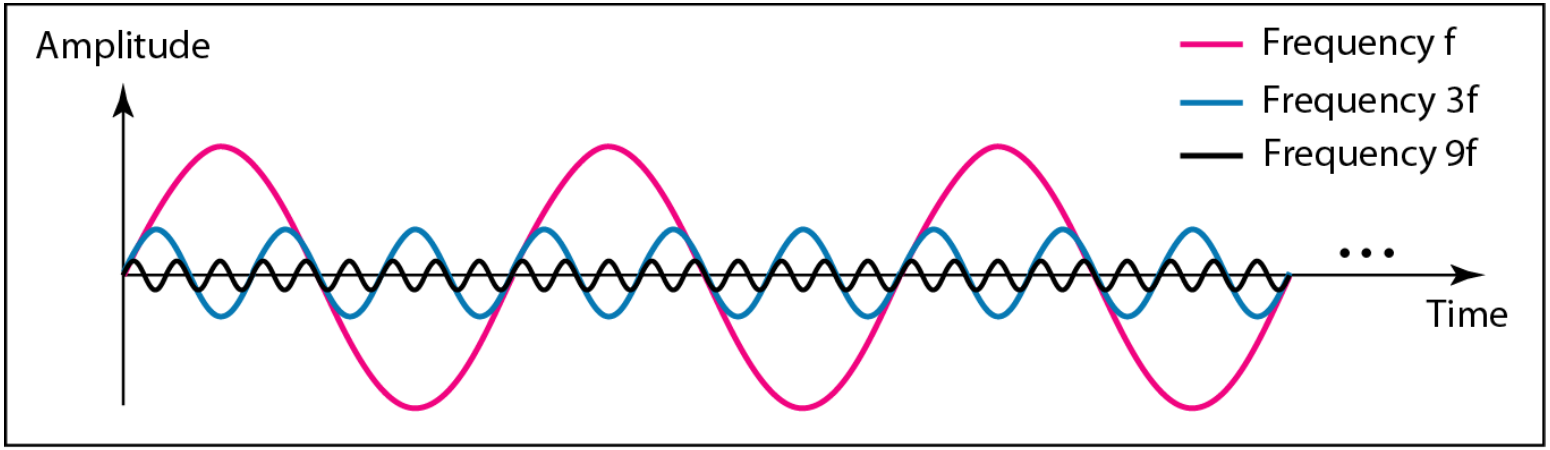
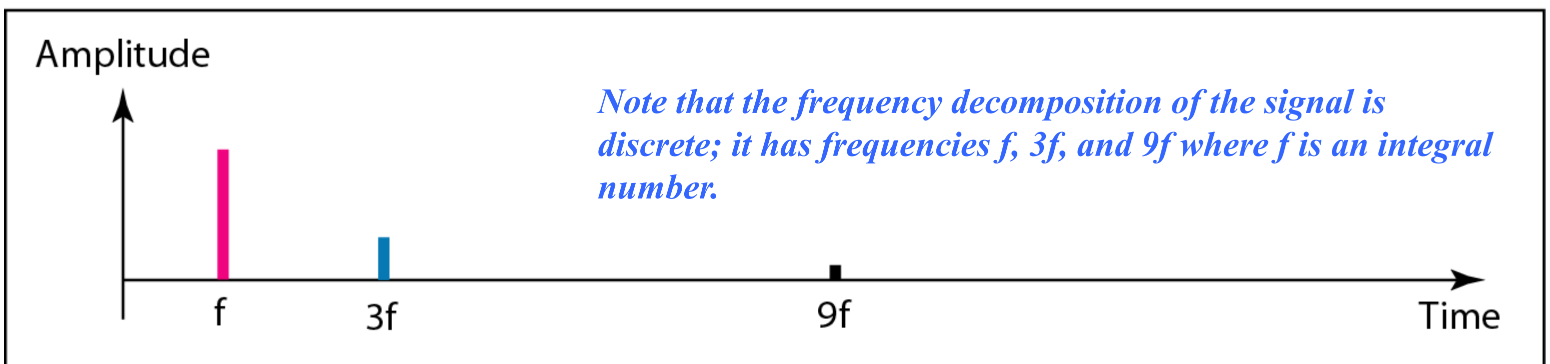


Figure 3.10 *Decomposition of a composite periodic signal in the time and frequency domains*



a. Time-domain decomposition of a composite signal

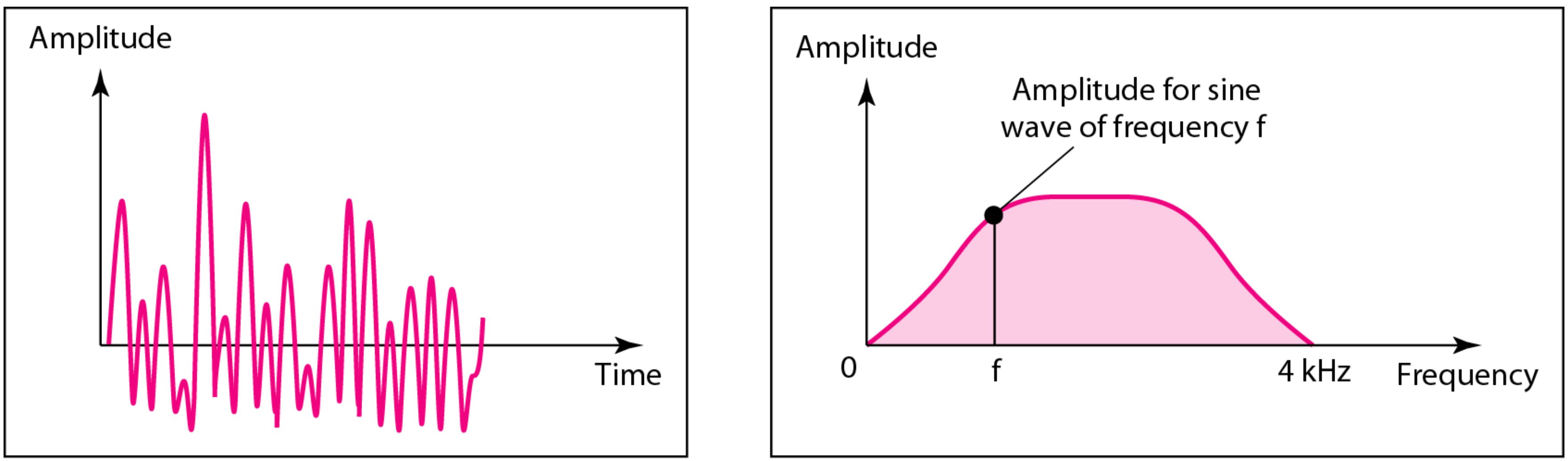


b. Frequency-domain decomposition of the composite signal

Example 3.5

Figure 3.11 shows a nonperiodic composite signal. It can be the signal created by a microphone or a telephone set when a word or two is pronounced. In this case, the composite signal cannot be periodic, because that implies that we are repeating the same word or words with exactly the same tone.

Figure 3.11 *The time and frequency domains of a nonperiodic signal*



a. Time domain

b. Frequency domain

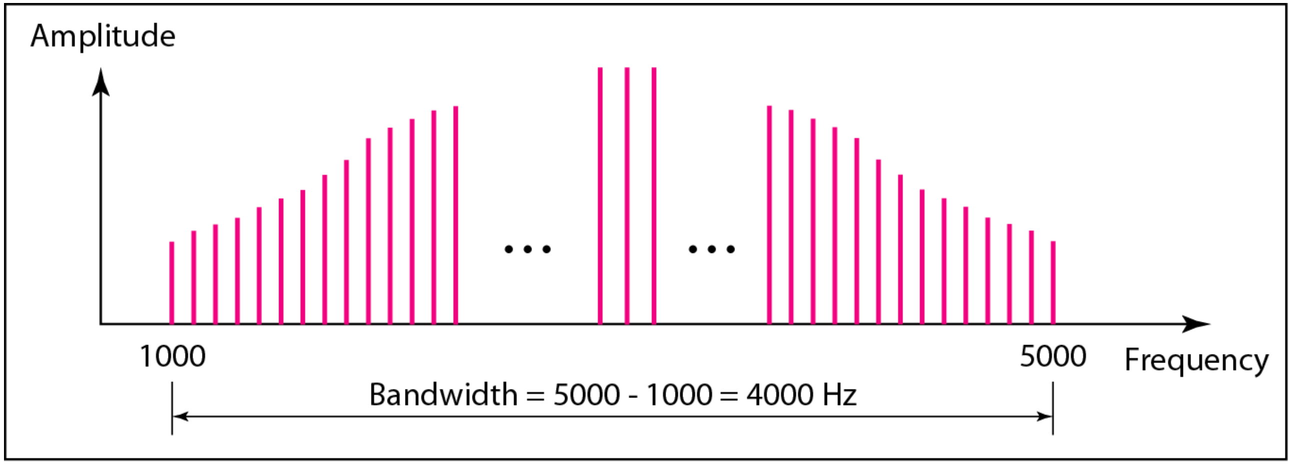
Composite Signals and Periodicity

- In a **time-domain** representation of this composite signal(speach), there are an **infinite number of simple sine frequencies**.
- Although the number of frequencies in a human voice is infinite, the **range is limited**.
- A normal human being can create a continuous range of frequencies between 0 and 4 kHz.
- Note that the frequency decomposition of the signal yields a **continuous curve**. There are an infinite number of frequencies between 0.0 and 4000.0 (real values)

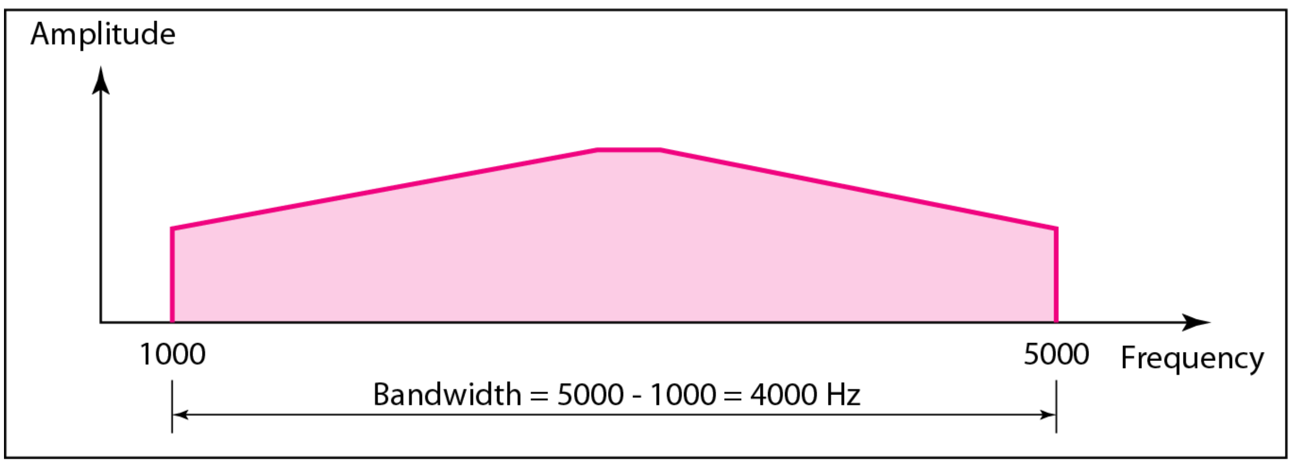
Bandwidth and Signal Frequency

- The **bandwidth** of a composite signal is the **difference between the highest and the lowest frequencies contained in that signal**.
- Figure 3.12 shows the concept of bandwidth.
- The bandwidth of the **periodic signal** contains all **integer frequencies** between 1000 and 5000.
- The bandwidth of the **nonperiodic** signals has the same range, but the **frequencies are continuous.**

Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example 3.6

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

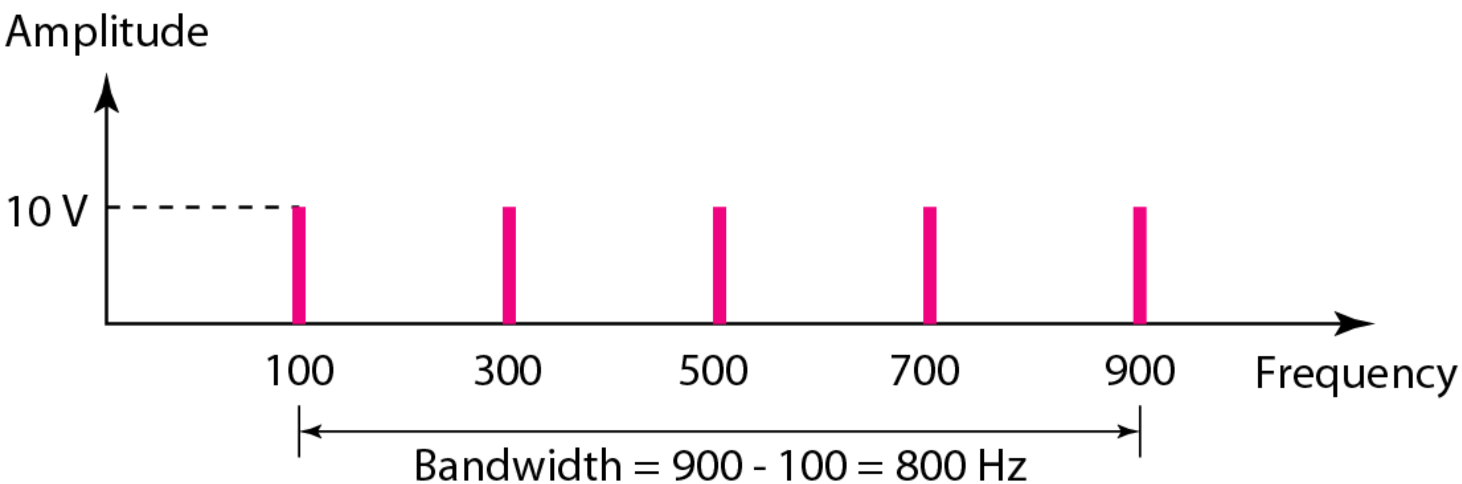
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$\boxed{B = f_h - f_l = 900 - 100 = 800 \text{ Hz}}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see Figure 3.13).

Figure 3.13 *The bandwidth for Example 3.6*



Example 3.7

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

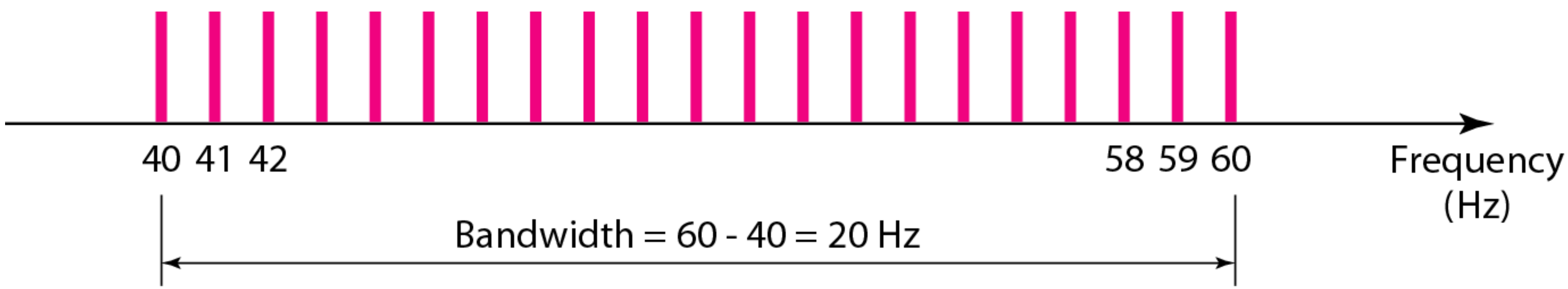
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes (see Figure 3.14).

Figure 3.14 *The bandwidth for Example 3.7*



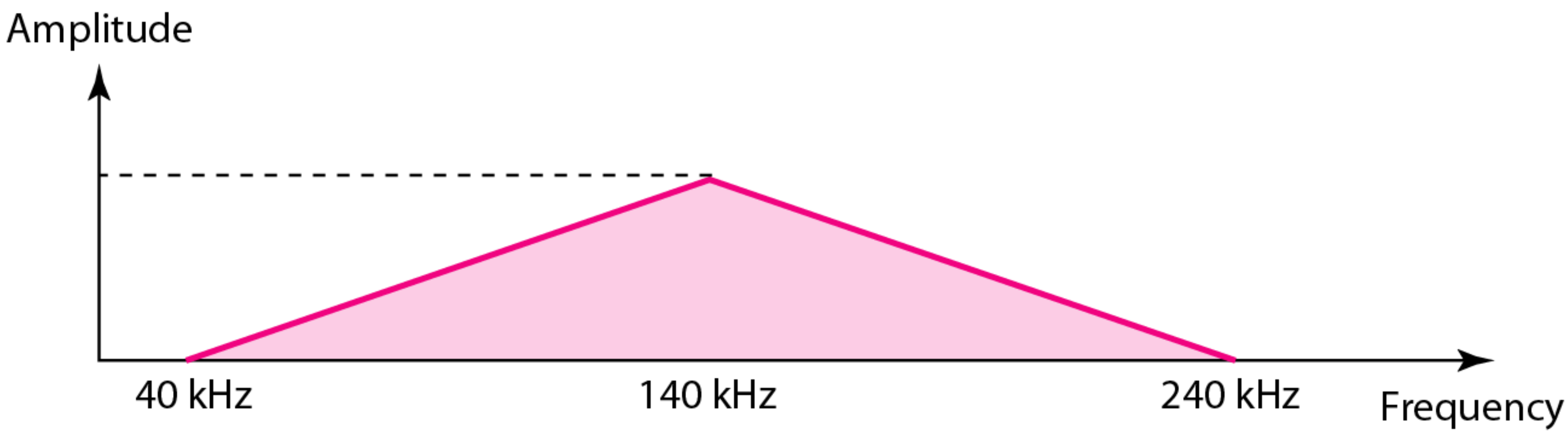
Example 3.8

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz. Figure 3.15 shows the frequency domain and the bandwidth.

Figure 3.15 *The bandwidth for Example 3.8*



Example 3.11

Another example of a nonperiodic composite signal is the signal received by an old-fashioned analog black-and-white TV. A TV screen is made up of pixels. If we assume a resolution of 525 × 700, we have 367,500 pixels per screen. If we scan the screen 30 times per second, this is 367,500 × 30 = 11,025,000 pixels per second. The worst-case scenario is alternating black and white pixels. We can send 2 pixels per cycle. Therefore, we need 11,025,000 / 2 = 5,512,500 cycles per second, or Hz. The bandwidth needed is 5.5125 MHz.

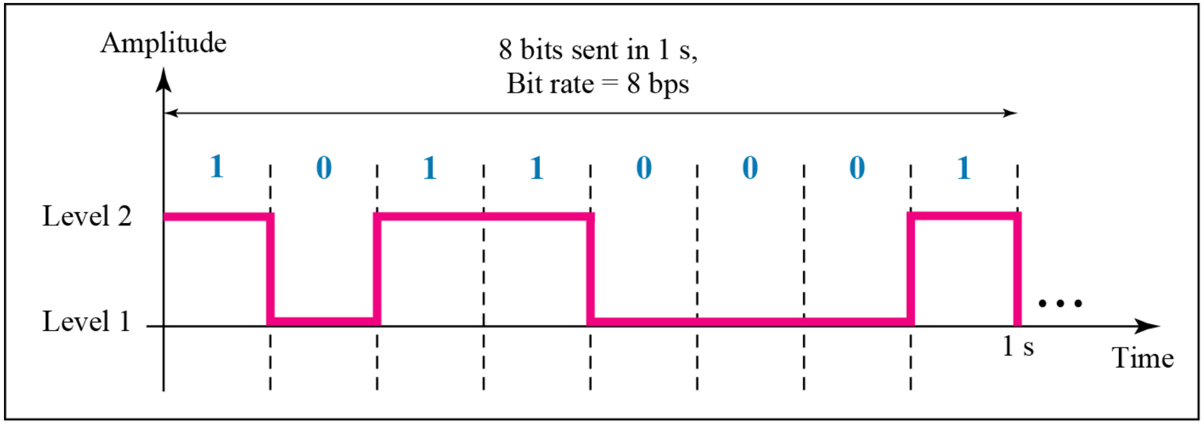
3-3 DIGITAL SIGNALS

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive(+5) voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

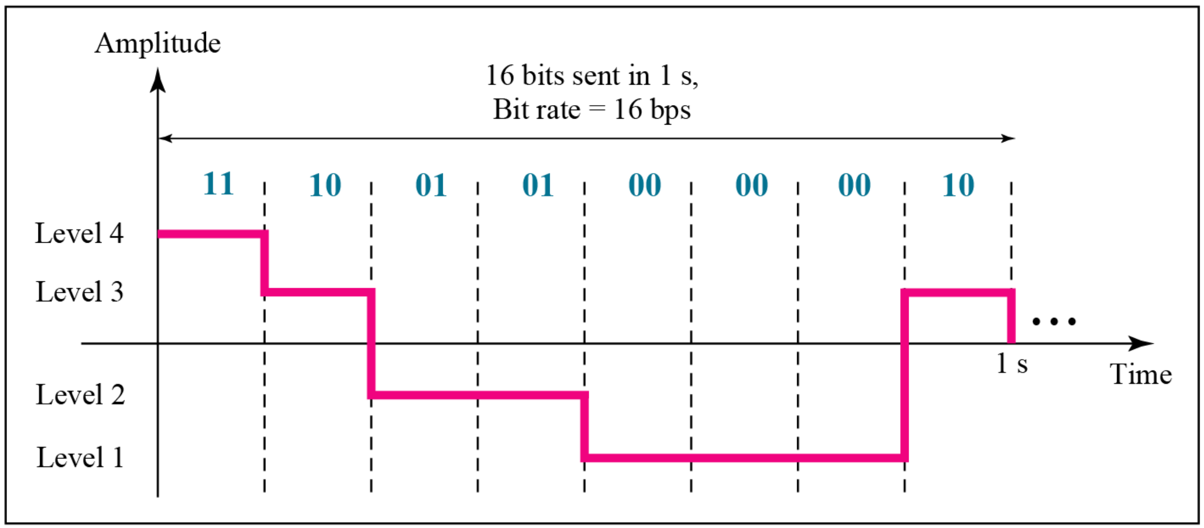
Topics discussed in this section:

- **Bit Rate**
- **Bit Length**
- **Digital Signal as a Composite Analog Signal**
- **Application Layer**

Figure 3.16 *Two digital signals: one with two signal levels and the other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels

Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\boxed{\text{Number of bits per level} = \log_2 8 = 3}$$

Each signal level is represented by 3 bits.

Example 3.17

A digital signal has nine levels. How many bits are needed per level?

We calculate the number of bits by using the formula in the previous example. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Bit Rate

- Most digital signals are nonperiodic, and thus period and frequency are NOT appropriate characteristics.
- Another term—bit rate (instead of frequency)—is used to describe digital signals.
- The **bit rate** is *the number of bits sent in 1s, expressed in bits per second (bps).*

Example 3.18

Assume we need to download text documents at the rate of 100 pages per sec. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits (ASCII), the bit rate is

$$\boxed{100 \times 24 \times 80 \times 8 = 1{,}636{,}000 \text{ bps} = 1.636 \text{ Mbps}}$$

Example 3.19

A digitized voice channel, as we will see in Chapter 4, is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

The bit rate can be calculated as

$$\boxed{2 \times 4000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}}$$

Example 3.20

What is the bit rate for high-definition TV (HDTV)?

Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

$$1920 \times 1080 \times 30 \times 24 = 1{,}492{,}992{,}000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

Bit Length

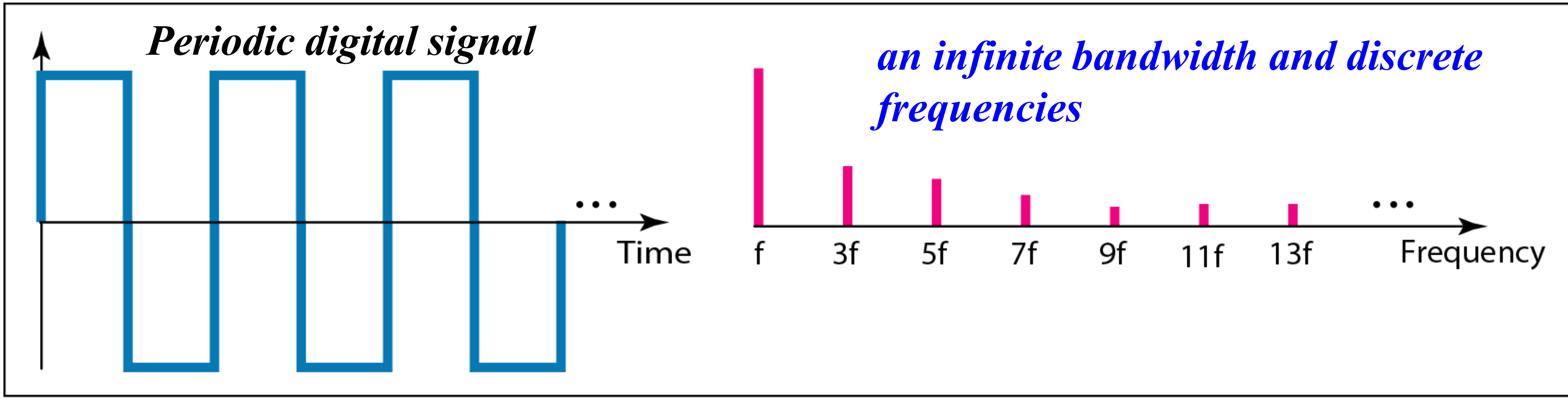
- We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium.
- We can define something similar for a digital signal: the bit length. The **bit length** is *the distance one bit occupies on the transmission medium.*

Bit length = propagation speed X bit duration

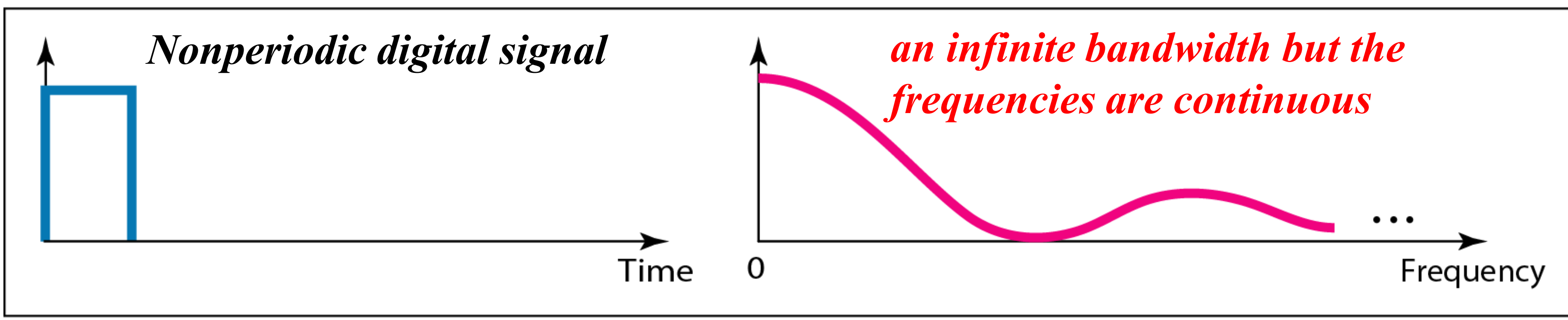
Digital Signal as a Composite Analog Signal

- Based on Fourier analysis, a digital signal is a composite analog signal.
- The bandwidth is infinite. We can intuitively come up with this concept when we consider a digital signal.
- A digital signal, in the time domain, comprises connected vertical and horizontal line segments.
- **A vertical line** in the time domain means **a frequency of infinity** (sudden change in time); **a horizontal line** in the time domain means **a frequency of zero** (no change in time).
- Going from a frequency of zero to a frequency of infinity (and vice versa) implies all frequencies in between are part of the domain. i.e., infinite bandwidth.

Figure 3.17 *The time and frequency domains of periodic and nonperiodic digital signals*



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal